

Integrating MLOps in Travel Productionization of ML Systems

Project Overview

This project demonstrates an end-to-end MLOps pipeline for automating the travel planning process using Machine Learning. The solution integrates flight price prediction, hotel recommendation, tourist attraction discovery, and user classification into a production-ready system. It combines machine learning model development with deployment, experiment tracking, REST APIs, containerization, and an interactive web application.

The project follows the complete machine learning lifecycle, including data preprocessing, feature engineering, model training, evaluation, deployment, model versioning, and user interaction through a Streamlit application.

Project Objectives

Predict flight prices using historical travel data. Recommend hotels based on user preferences and travel history. Discover tourist attractions for travel destinations. Classify user gender using machine learning techniques. Develop production-ready REST APIs using Flask. Track experiments and model versions using MLflow. Containerize the application using Docker. Build an interactive Streamlit dashboard for visualization and recommendations. Demonstrate an end-to-end MLOps workflow for travel analytics.

Data Preprocessing

The following preprocessing techniques were applied:

Missing value handling Duplicate removal Date conversion Feature encoding Numerical feature normalization Feature scaling Feature engineering Dataset merging Data validation

Feature Engineering

Flight Features Travel Month Travel Day Travel Weekday Weekend Indicator Flight Distance Travel Duration Flight Type Airline Agency Hotel Features Average Hotel Cost Price Per Day Popularity Score Travel Frequency Previous Hotels Destination User Features Age Group Gender Company Travel Count

Machine Learning Models

Flight Price Prediction (Regression)

Implemented models:

Linear Regression Random Forest Regressor Gradient Boosting Regressor XGBoost Regressor CatBoost Regressor LightGBM Regressor Evaluation Metrics Mean Absolute Error (MAE) Root Mean Squared Error (RMSE) R^2 Score Mean Absolute Percentage Error (MAPE) Gender Classification

Business Use Case 1: Flight Price Prediction

Business Problem

Flight ticket prices vary depending on several factors, including travel distance, travel duration, airline agency, travel date, departure city, destination, and flight type. Customers often find it difficult to determine whether the current ticket price is reasonable.

Business Insight

Analysis of the Flights dataset indicates that flight prices are influenced by multiple travel-related factors rather than a single variable. Feature engineering using travel month, weekday, and weekend indicators improves the model's ability to capture seasonal and temporal travel patterns. Regression models provide estimated ticket prices that support informed travel planning.

Hotel Recommendation

The recommendation engine combines:

- Content-Based Filtering
- Collaborative Filtering
- Popularity-Based Recommendation

This hybrid approach improves recommendation quality by considering destination characteristics, hotel attributes, historical bookings, and user preferences.

Business Use Case 2: Hotel Recommendation

Business Problem

Travelers often spend significant time searching for hotels that match their destination, budget, and preferences.

Business Insight

The Hotels dataset enables personalized recommendations using hotel location, booking price, stay duration, and historical booking information. Combining content-based filtering, collaborative filtering, and popularity-based recommendations increases the relevance of suggested hotels.

Gender Classification

The following classification algorithms were compared:

- Logistic Regression

- Random Forest
- XGBoost
- LightGBM

Evaluation metrics included:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

The model with the strongest overall evaluation metrics was selected for deployment.

Business Use Case 3: User Analytics

Business Problem

Travel companies require user insights to understand customer characteristics and improve service personalization.

Business Insight

The Users dataset enables gender classification using demographic features such as company and age. The classification model supports user segmentation for analytical purposes and can contribute to personalized marketing strategies, provided it is used responsibly and in accordance with privacy and fairness considerations.

MLflow

MLflow is used for:

- Experiment Tracking
- Parameter Logging
- Metric Logging
- Artifact Management
- Model Versioning
- Model Registry

Tracked models include:

- Random Forest
- XGBoost
- LightGBM
- CatBoost

Streamlit Dashboard

The Streamlit application provides:

- Flight Price Prediction
- Hotel Recommendation
- Travel Analytics

Conclusion

The project demonstrates how machine learning and MLOps can be combined to build a production-ready travel analytics platform. By integrating flight price prediction, hotel recommendation, user analytics, and scalable deployment practices, the solution supports more efficient travel planning and data-driven decision-making. The use of MLflow, Flask, Docker, and Streamlit ensures that the models are reproducible, deployable, and maintainable, making the platform suitable as a foundation for future enhancements such as real-time booking integration, live pricing updates, and expanded destination recommendations.